COMP 4107: Neural Networks – Assignment 2 – Winter 2026 –

Due: February 13th, 2026

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Question 1

Please see `assignment2.py` for implementation.

Question 2

Please see `assignment2.py` for implementation.

Question 3

Please see `assignment2.py` for implementation.

Question 4

Please see `assignment2.py` for the full implementation.
The following plots were generated for the below experiments. Note that the data was split into 70% training data, 15% validation data, and 15% test data for all experiments.

(a) Number of neurons in each hidden layer

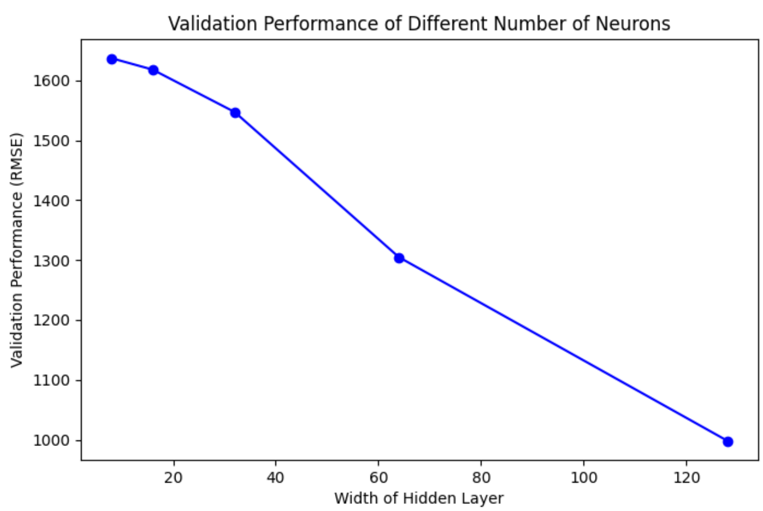

Figure 1: Neurons Experiments

(b) Number of hidden layers in the network

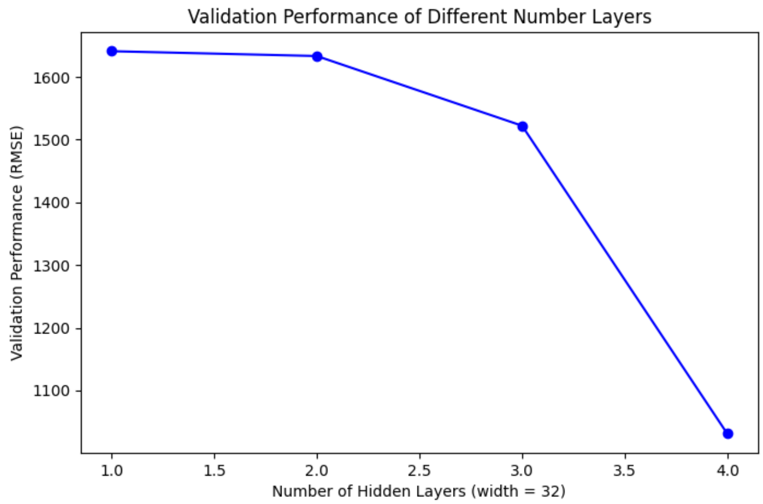

Figure 2: Layers Experiments

(c) Number of epochs used for training

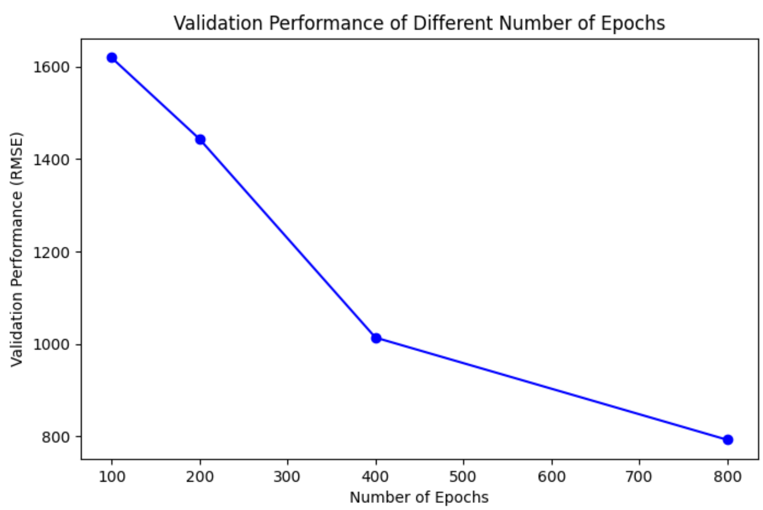

Figure 3: Epochs Experiments

(d) Different activation functions

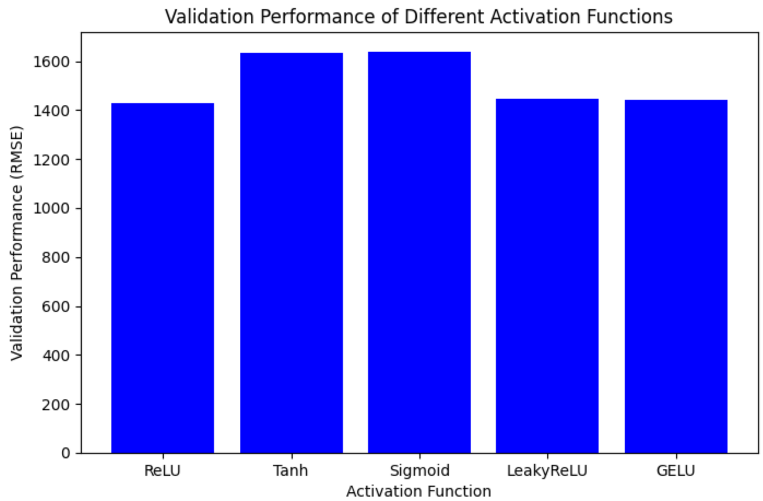

Figure 4: Activation Function Experiments

(e) Model with the best performance
The best single performing model out of all experiments was the epoch model in part (b) that used 800 epochs. The performance on the training, validation, and test set is given below.

Best Model Performance (single model from experiments): epochs = 800

- Train RMSE: 744.9811401367188
- Validation RMSE: 822.7666625976562
- Test RMSE: 800.5404052734375

The best overall performing model when combining the models from each experiment was the following model.
Best Overall Model Performance (combining models from experiments):

- Neurons: 128
- Layers: 3 layers, each with width = 32 ([32, 32, 32])
- Epochs: 800
- Activation Function: ReLU()

The performance on the training, validation, and test set is given below.

- Train RMSE: 616.2930297851562
- Validation RMSE: 639.0341796875
- Test RMSE: 704.6664428710938

We can see that combining the best models from each experiment yields a lower RMSE compared to the single model.